

A fixed-direction Solar-System Lorentz-violation target: the $\bar{s}^{\text{TX}}$ boost-dipole of de Sitter–Unruh modified inertia

Carl P. Zimmerman · Briar Creek Tech · 2026-06-27 A prediction note for the planetary-ephemeris community (INPOP / EPM / DE).

The ask, in one line

The de Sitter–Unruh modified-inertia account of the MOND scale is a **preferred-frame** theory, so it induces a **specific, forced, non-zero gravity-sector Lorentz-violation coefficient** with a **fixed sky direction**. A dedicated ephemeris fit for the $\bar{s}^{\text{TX}}$ **boost-dipole** at the $\sim 9 \times 10^{-10}$ level — a factor ≈ 1.5 below the current bound — would **detect or exclude** it. The data already exist; the dedicated dipole fit has, to our knowledge, not been run.

Why there is a forced $\bar{s}^{\text{TX}}$

If inertia is a body’s response to the de Sitter–Unruh bath of the cosmological horizon (Milgrom 1999; Deser–Levin 1997), the bath defines a **preferred rest frame — the cosmic (CMB) frame**. A preferred frame *is* a Standard-Model-Extension (SME) gravity-sector background $\bar{s}_{\mu\nu}$ (Bailey & Kostelecký 2006; Kostelecký & Tasson 2011). The Solar System moves at $\mathbf{v} = 369.8 \text{ km s}^{-1}$ toward the CMB apex, $\boldsymbol{\beta} = \mathbf{v}/c = 1.23 \times 10^{-3}$, so the otherwise-isotropic background acquires a **time–space (boost) component** $\bar{s}^{\text{TX}} \propto \boldsymbol{\beta}$ aligned with that apex. Its magnitude is inherited from the same $a_0 = cH_0/Z = 9.36 \times 10^{-11} \text{ m s}^{-2}$ that sets the MOND scale (component ledger in the author’s published SME-bridge work).

The prediction (specific and falsifiable)

quantity	value
coefficient	$\bar{s}^{\text{TX}}$ (gravity sector, boost-dipole)
predicted magnitude	$\approx 8.7 \times 10^{-10}$
sign/CPT	CPT-even only (the CPT-odd k_{AF} channel vanishes)
sky direction (fixed, not fit)	the CMB apex: (l, b) = (264.0°, 48.3°); (RA, Dec) $\approx$ (168°, −7°)
current bound (ephemerides)	$ \bar{s}^{\text{TX}} \lesssim 1.3 \times 10^{-9}$ (Hees et al. 2015)
margin	$\approx 1.5\times$ under the present limit

The fixed direction is the discriminating feature: this is **not** a free coefficient to be marginalized, but a dipole **locked to a known point on the sky**, so the fit has one fewer degree of freedom than a generic SME search.

The decisive measurement

A dedicated $\bar{s}^{\wedge}\text{TX}$ **dipole fit** — orbital dynamics plus light propagation in the Time-Transfer Formalism (Hees et al. 2015) — reaching $\sigma(\bar{s}^{\wedge}\text{TX}) \approx 4 \times 10^{-10}$ (a factor ≈ 1.5 –2 beyond the present sensitivity) will:

- **detect** a dipole of $\approx 8.7 \times 10^{-10}$ aligned with the CMB apex $\rightarrow$ support for a preferred-frame (modified-inertia) origin of the MOND scale; or
- **bound it below** $8.7 \times 10^{-10} \rightarrow$ exclude this preferred-frame realization.

The data are in hand: INPOP / EPM / DE planetary ranging, the Cassini Saturn arc (extended), Mars-orbiter ranging, and **BepiColombo** Mercury ranging (arrival 2026) tighten exactly the channels that carry $\bar{s}^{\wedge}\text{TX}$. This is **analysis-limited, not data-limited**.

Honest scope (what this is and is not)

- The magnitude inherits the framework’s own modeling of the $a_0 \rightarrow \bar{s}_{\mu\nu}$ map; the **direction and CPT-even character are robust** consequences of “preferred frame = CMB frame,” the **magnitude is framework-specific** ($\pm O(1)$).
- This is a test of the **gravity-sector preferred frame only** — it is a theory of **gravity and the dark sector, not a theory of everything**; the Standard-Model matter sector is untouched.
- The framework **passes every other named gravity-sector bound** (LLR, INPOP/Cassini γ , atom interferometry, pulsar α_z , VLBI); $\bar{s}^{\wedge}\text{TX}$ is its **tightest** exposure, which is why it is the one most worth a dedicated fit.

References

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Reproducible: reviews/stx_target.py. Footing $a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}$, $Z = \sqrt{(32\pi/3)}$.